

Customer Shopping Behavior Analysis

1. project Overview

This project examines the customer shopping behavior by using transactional data of 3900 purchases in different product categories. The objective is to get insights on spending, customer segment, product preference, and subscription behavior that would help in strategic business decisions.

2. Dataset Summary

- rows: 3900
- columns = 18
- Key Features=Demographics of Customers (Age, Gender, Location, Subscription Status) Key Features= Demographics of Customers (Age, Gender, Location, Subscription Status)
purchase details (item bought, category, purchase amount, season, size, colour)
shopping bahavior(discount applied, promo code used, previous purchases, frequency of purchase, review rating, shipping type)
- Missing data = 37 values in review rating columns

3. Exploratory Data analysis using python

Missing data: checked null values and inputted missing values in review Rating columns with median rating of each_product catagory

Feature engineering: feature engineering:

- Bin customer ages into age groups by columns.
- created purchase_frequency_days column from purchase data

Database integration: linked python script to mySQL and uploaded cleaned DataFram into database for sql analysis

4. Data Analysis Using SQL

we perfomed structured analysis in mySQL to answer key business quetions:

1.Total Revenue ganrated by male vs female customers

	gender	revenue
▶	Male	157890
	Female	75191

2. Which customer used discount but spend more money then avrage customer purchsed amount

	customer_id	purchase_amount
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68

- 3.Top five best review product

	item_purchased	avr
▶	Gloves	3.8614285714285725
	Sandals	3.8443750000000003
	Boots	3.8187500000000005
	Hat	3.8012987012987005
	Skirt	3.784810126582278

4. Compare the avarage purchase amount between standard and express shpping

	shipping_type	avg(purchase_amount)
▶	Express	60.4752
	Standard	58.4602

5. compare subscribe and non subscribe

	subscription_status	customer	avg_spend	revenue
▶	Yes	1053	59.4919	62645
	No	2847	59.8651	170436

6. Segment customer into new , returning and loyal purchase show the count of each segment

	customer_segment	number of customer
▶	loyal	3116
	returning	701
	new	83

7. the customers who are repate buyers also likely to subscribe

	subscription_status	buyer
►	Yes	958
	No	2518

8. What is the revenue contribution by age

	age	revenue
►	67	3258
	44	3312
	20	3424
	48	3576
	60	3730
	34	3759

Dashboard in Power BI

Finally , we built an interactive dashboard in **Power BI**

